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SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
13 20 21	13 20 21	13 20 21	13 20 21	13 20 21	13 20 21	13 20 21	13 20 21	13 20 21	13 20 21
understanding 2.25	2	2.25	2.25	2	1	2	2.25	2.25	2
45%									

Explanation	1	0.5	1	1.75	1	0.5	1	1.75	1
35%									

Clarity	0.5	0.5	0.5	1	0.5	0.5	0.5	1	0.5
20%									
	3.75	3	3.75	5	3.5	2	3.5	5	3.5

38
50

76%

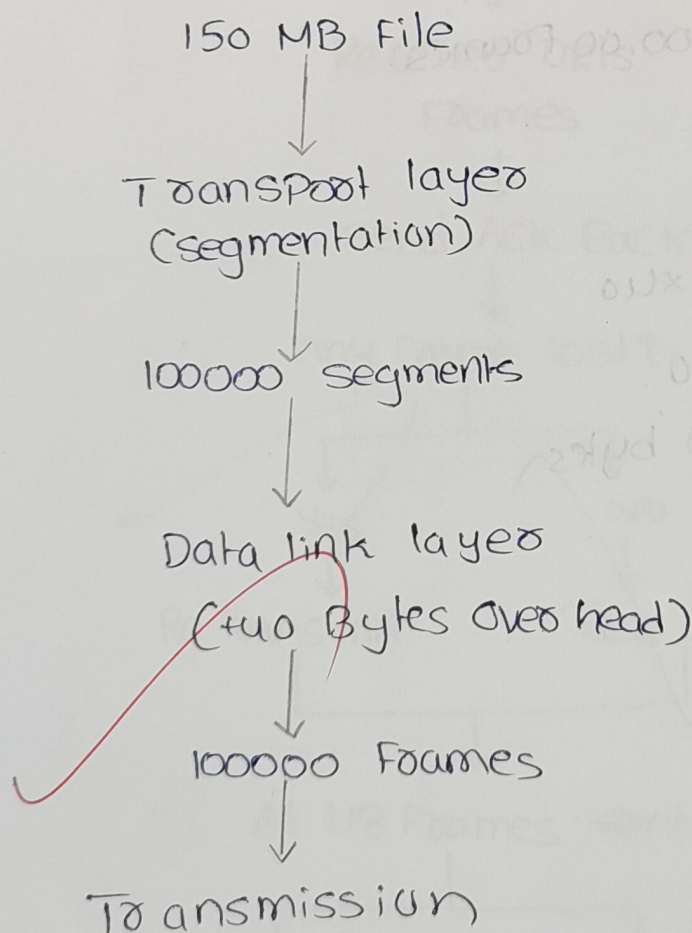
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1A) Design File Transfer

Transport layer and Data link layer Communication:

* The Transport layer divides the files into segments and ensures reliable delivery. The Data link layer converts segments into frames and adds headers and trailers for error detection and node-to-node communication.

Flow Chart:



Calculation:

Given:

* File size = 150 MB = 150,000,000 bytes

Number of Check Points:

$$\begin{aligned}\text{checkPoints} &= 340 \div 90 \\ &= 6.5 \\ &= 6 \text{ check Points}\end{aligned}$$

Data Retransmitted:

$$\begin{aligned}\text{last check Point} &= 6 \times 60 \\ &= 360 \text{ MB}\end{aligned}$$

Retransmitted Data

$$\begin{aligned}&= 340 - 360 \\ &= 30 \text{ MB}\end{aligned}$$

Result:

* Check Points Created = 6

* Retransmitted Data = 30 MB

GA) Compression and Encryption:

* The Presentation layer compresses data to reduce transmission size and encrypts data to provide security and confidentiality.

Flow Chart:

900 MB File



Compression
(40%)



Encryption → 567 MB → Transmission
(+5%)

IP Fragmentation:

Fragmentation in Network layer:

- * The Network layer performs routing, fragmentation and reassembly of packets to ensure successful delivery.

Flow Chart:

12000 Byte Packet



Check MTU,
(1500 Bytes)



Fragmentation



q Fragments created



Transmission



Reassembly at Destination

Calculation:

Given:

* Packet size = 12,000 bytes

* MTU = 1500 bytes

* IP Header = 20 bytes

Calculate Data Per Fragment

$$\underline{\text{Data Per Fragment}} = \text{MTU} - \text{Header size}$$

Calculation:

- * Application = 4ms
- * Transport = 3ms
- * Network = 5ms
- * Data link = 2ms
- * Physical = 1ms
- * Transmission = 19ms
- * Propagation = 18ms

Total Delay:

Total Delay = Processing + Transmission + Propagation.

$$\begin{aligned}\text{Total Delay} &= 4 + 3 + 5 + 2 + 1 + 19 + 18 \\ &= 62\text{ms}\end{aligned}$$

Result:

* ~~Total end to end Delay~~ = 62ms

Q1) Protocol overhead:

Protocol overhead consists of additional control information added by OSI layers. Excessive overhead reduces network efficiency.

Flow Chart:

80 MB Data



Frame creation
(+80 Bytes)



52632 Frames → Transmission.

$$\text{Data Per Fragment} = 1500 - 20$$

$$= 1480 \text{ bytes}$$

Calculate Number of Fragments:

$$\text{Number of Fragments} = \text{Packet Size} \div \text{Data Per Fragment}$$

$$\text{Number of Fragments} = \text{Packet Size} \div \text{Data Per Fragment}$$

$$= 12000 \div 1480$$

$$= 8.108$$

$$\approx 9 \text{ Fragments}$$

Calculate Header over head:

$$\text{Header over head} = \text{Number of Fragments} \times \text{Header size}$$

$$\text{Header over head} = 9 \times 20$$

$$= 180 \text{ bytes}$$

Result:

$$\ast \text{Data Per Fragment} = 1480 \text{ bytes}$$

$$\ast \text{Number of Fragments} = 9$$

$$\ast \text{Header over head} = 180 \text{ bytes}$$

4A) Sliding window Protocol

* The sliding window Protocol allows multiple frames to be transmitted before receiving acknowledgements, improving throughput and efficiency

Transmission Time:

$$\text{Time} = (\text{Data} \times 8) \div \text{Throughput}$$

$$\text{Time} = (3000 \times 8) \div 120$$

$$= 24000 \div 120$$

$$\text{Time} = 200$$

Result:

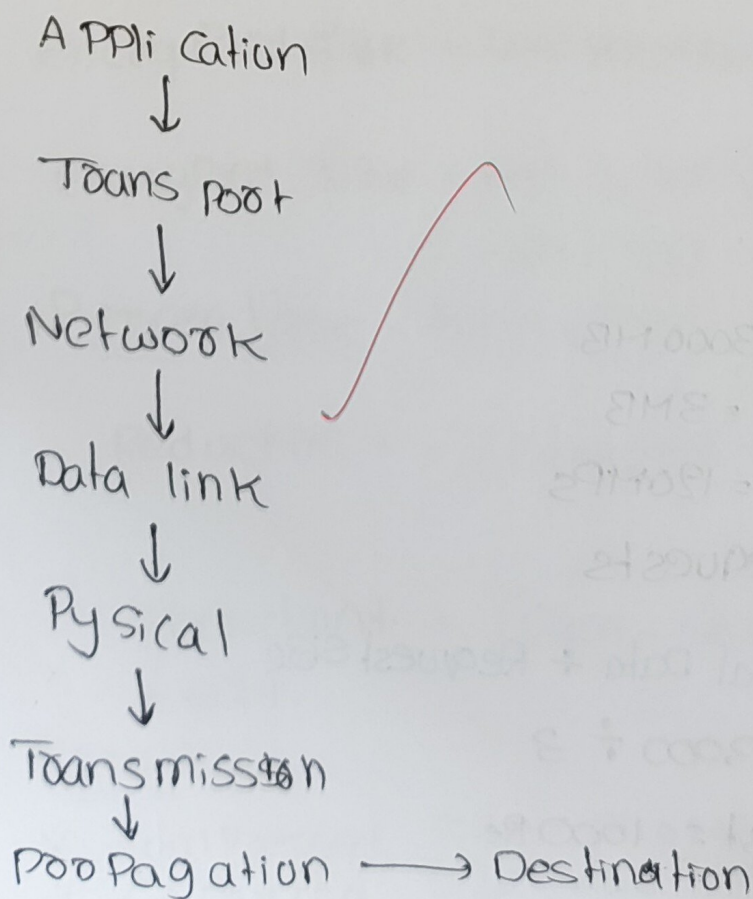
* Request = 1000

* Transmission Time = 200s

8A) End to end Delay:

* Each OSI layer introduces Processing delay. Total delay is the sum of Processing, transmission, and Propagation delays.

Flow chart:



* Reduction

* Segment size = 1500 bytes

* overhead = 40 bytes / frame

* Band width = 50 Mbps

Number of segments:

$$\text{Number of segments} = \frac{\text{File size}}{\text{segment size}}$$

$$\begin{aligned}\text{Number of segments} &= \frac{150,000,000}{1500} \\ &= 100,000 \text{ segments}\end{aligned}$$

Number of Frames:

* Num of Frames = 100,00 Frames

Total overhead

$$\text{over head} = \text{Frames} \times 40$$

$$= 100,000 \times 40$$

$$= 4,000,000 \text{ bytes}$$

$$= 4 \text{ MB}$$

Transmission Time:

$$\text{Time} = \frac{154 \times 8}{50}$$

$$\text{Time} = 24.64 \text{ s}$$

Result:

* segments = 100,000

* Frames = 100,000

* overhead = 4 MB

* Transmission Time = 24.64 s

Compressed size:

$$= 240 \times 70 / 100$$

$$= 168 \text{ MB}$$

Number of segments:

$$= 168,000,000 \div 1000$$

$$= 168,000 \text{ segments}$$

Total overhead:

$$= 168,000 \times 60$$

$$= 7,200,000 \text{ bytes}$$

$$= 7.2 \text{ MB}$$

Final Data Transmitted

$$= 168 + 7.2$$

$$= 175.2 \text{ MB}$$

Result:

$$\times \text{Compressed size} = 168 \text{ MB}$$

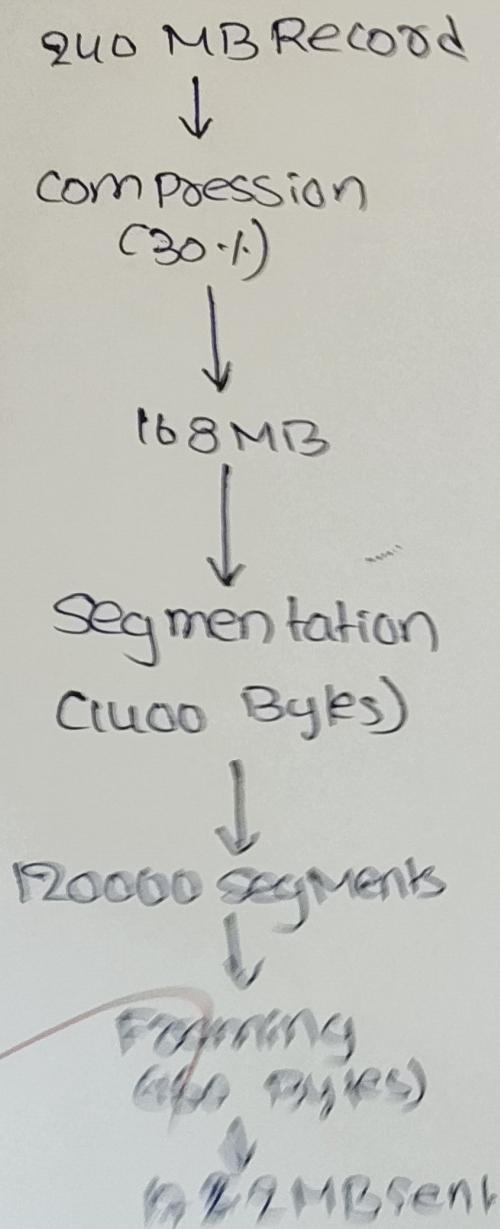
$$\times \text{segments} = 168,000$$

$$\times \text{overhead} = 7.2 \text{ MB}$$

$$\times \text{Final Data} = 175.2 \text{ MB}$$

Multiple OSI layers work together. The Transport layer compresses data, the Transport layer segments and the Data link layer adds framing overhead.

Flow Chart:



Calculation:

Given:

- * Initial Data = 240 MB
- * Compression Ratio = 30:1
- * Segment Size = 1400 bytes
- * Overhead = 60 bytes / Frame.

Result:

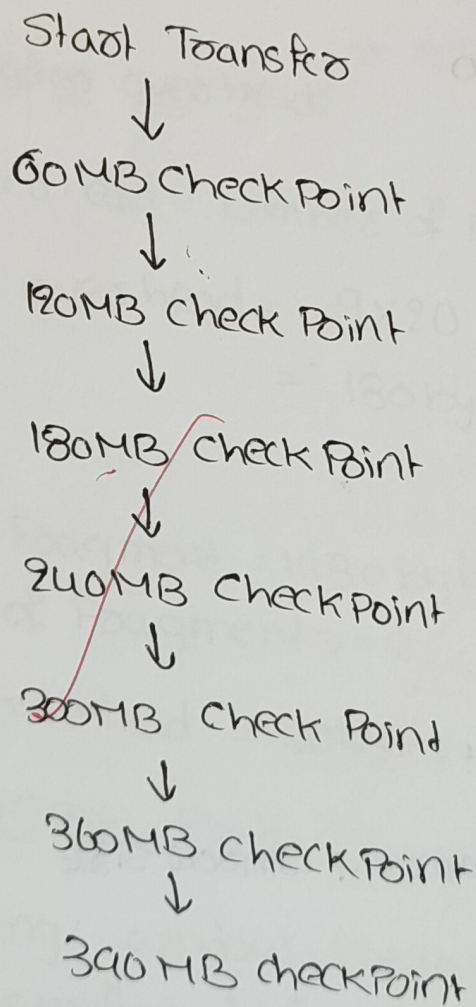
* windows = 8

* Retransmission = 8

5A) session layer synchronization:

* The session layer inserts synchronization check points that allow communication to resume from the last check point after a failure.

Flow chart:



Calculation:

Given:

* Video size = 600 MB

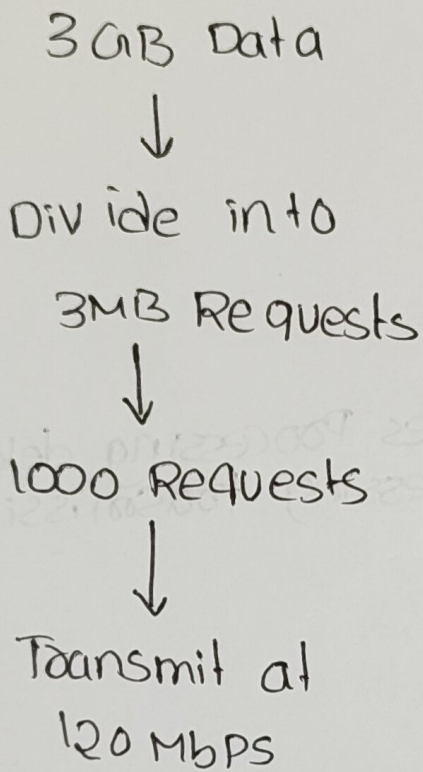
* Check Point = Every 60MB

* Failure = 390MB

FTP APPLICATION

* The Application layer provides services such as FTP, HTTP, and email. FTP enables reliable file transfer between systems.

Flow chart:



Calculation:

Given:

* Data = 3GB = 3000 MB

* Request size = 3MB

* Through Put = 120Mbps

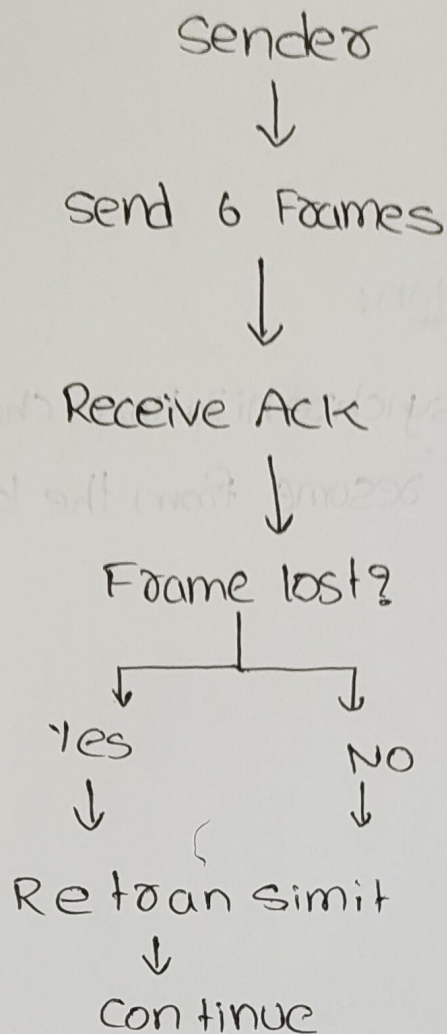
Number of Requests

Requests = Total Data ÷ Request Size

Requests = $3000 \div 3$

Request = 1000 R

Flow chart:



Calculation:

Given:

* window size = 6

* Total Frame = 48

Number of windows:

$$\text{windows} = \text{total Frames} \div \text{window size}$$

$$\text{windows} = 48 \div 6$$

$$\text{windows} = 8$$

Retran smi ssion:

$$\text{Retran smission} = 8 \times 1$$

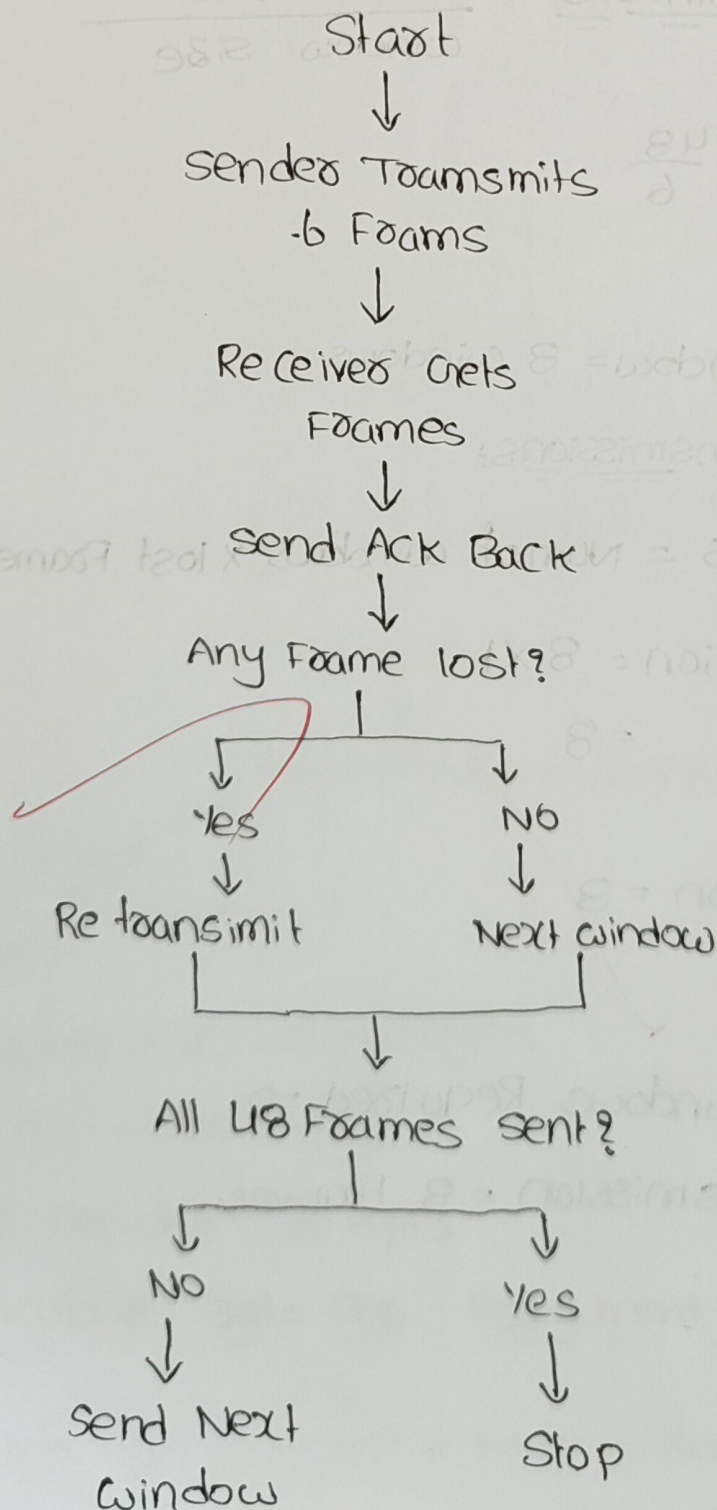
$$= 8 \text{ Frames.}$$

Sliding window Protocol

Flow control in Transport layer:

* Flow control Prevents the sender from transmitting data faster than the receiver can process; improving communication efficiency.

Flow Chart:



Calculation:

Given:

* window size = 6 Frames

* Total Frames = 48 Frames

* one frame lost in each window

Calculate Number of windows:

$$\text{Number of windows} = \frac{\text{Total Frames}}{\text{window size}}$$

$$\text{substitution:} = \frac{48}{6}$$

Result:

Number of window = 8 windows

Calculate Retransmissions:

Retransmissions = Num of windows x lost Frames per window

$$\text{Retransmission} = 8 \times 1$$
$$= 8$$

Result:

$$\text{Retransmission} = 8$$

Final Answers:

* Number of windows Required = 8

* total Retransmission = 8 Frames

Calculation:

Given:

* Useful Data = 80MB = 80,000,000 bytes

* Frame size = 1600 bytes

* over head = 80 bytes

Payload Per Frame:

$$\begin{aligned}\text{Payload Per Frame} &= 1600 - 80 \\ &= 1520 \text{ bytes}\end{aligned}$$

Number of Frames:

$$\begin{aligned}\text{Number of Frames} &= 80,000,000 \div 1520 \\ &= 52,632 \text{ Frames}\end{aligned}$$

Total over head:

$$\begin{aligned}\text{Total over head} &= 52,632 \times 80 \\ &= 4,210,560 \text{ bytes}\end{aligned}$$

Efficiency:

$$\begin{aligned}\text{Efficiency} &= (\text{Payload} \div \text{Frame size}) \times 100 \\ &= (1520 \div 1600) \times 100 \\ &= 95\%\end{aligned}$$

Result:

* Frames = 52,632

* over head = 4,210,560 bytes

* Efficiency = 95%.

Calculation:

Given:

* original File = 900 MB

* Compression = 40%.

* Encryption overhead = 5%.

Compressed File size:

Formula:

$$\text{Compressed size} = \text{original size} \times (100 - \text{compression}) / 100$$

$$\begin{aligned}\text{Compressed size} &= 900 \times 60 / 100 \\ &= 540 \text{ MB}\end{aligned}$$

Encrypted File size:

$$\text{Encrypted size} = \text{Compressed size} \times (100 + \text{overhead}) / 100$$

$$\begin{aligned}\text{Encrypted size} &= 540 \times 105 / 100 \\ &= 567 \text{ MB}\end{aligned}$$

Percentage Reduction:

$$\text{Reduction \%} = ((\text{original size} - \text{Final size}) / \text{original size}) \times 100$$

$$\begin{aligned}\text{Reduction \%} &= (900 - 567) / 900 \times 100 \\ &= 37\%\end{aligned}$$

Result:

- * Compressed size = 540 MB
- * Encrypted size = 567 MB
- * Reduction = 37%